Emulator Practice Guide

PRACTICE ONLY. The emulator is for learning to operate Hexis Forensics without a real device and with no risk to a case. It **is not a real device**: it has no real IMEI or baseband, the data partition is different, and it comes with **elevated privilege (root)**, which a common phone does not have. For that reason, **never** use the emulator to claim that an extraction will work on a real phone, nor as a basis for an expert report. It is a lab, not proof.

1. INSTALL ANDROID STUDIO AND CREATE THE EMULATOR

A. Android Studio and the right image

1. Download and install **Android Studio** (developer.android.com/studio). It is free and runs on Windows. Accept the standard components during installation.
2. Open the **Device Manager** (phone icon on the right) > **Create device** > pick a **Pixel** > **Next**.
3. **System Image:** pick **Android 13 (API 33)** in a version **without Google Play** — the label should read “Google APIs” or “AOSP”, **not** “Google Play”. The Google Play image **blocks root** and is less useful for practice. Download the image.
4. **Finish** to create the emulator. Give it a name that signals practice, e.g. **Practice-Bench**.

B. Start it and confirm elevated privilege (root)

5. In the Device Manager, click **Play** to start the emulator and wait for the Android home screen to open.
6. On the computer, open the Terminal in the **adb** folder (or with adb on the PATH) and type **adb devices**: it should list something like **emulator-5554 device**. The emulator comes authorized — there is no cable and no RSA fingerprint.
7. Type **adb root**: it should answer “restarting adbd as root”. That privilege — which a common phone does not have — is what lets the emulator run the modules that require root.

2. POPULATE TEST DATA

C. Contacts, SMS and WhatsApp

8. **Contacts:** open the emulator's Contacts app and create a few, with name and number.
9. **SMS:** on the emulator's side bar, **Extended controls** (the three-dots button “...”) > **Phone** > type a number and a message > **Send message**: a test SMS arrives on the emulator.
10. **WhatsApp:** download the **APK from the website** (whatsapp.com/android — there is no Play Store on the emulator) and **drag the .apk file onto the emulator window** to install. Set it up with a number whose SMS you can receive and exchange a few messages to have content.

Gmail does not run on the emulator without Google Play — this is expected, not a defect. Practice here covers contacts, SMS, media and WhatsApp, which is what the tool extracts on Android.

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3. EXTRACT WITH HEXIS

D. Connect and open a practice case (manual, ch. 3–4)

1. With the emulator open, in the tool: **Android** page > **Verificar Conexão** (check connection). The emulator shows up connected and authorized, with no cable.
2. **Iniciar Extração** (start extraction) > confirm the examiner > read **Método e limitações** > **Ciente — Prosseguir**.
3. **Case:** create a new case and make **practice** explicit in the name (e.g. PRACTICE-001). **Never mix it with a real case.** Fill device, seal and the photos with fictitious data.

E. Select the modules and extract

4. The common modules are already ticked. Since the emulator has **elevated privilege**, you may also tick the **express-selection** ones (those that alter the device / require root) to see how the tool runs and records them.
5. To practice WhatsApp, follow the **64-digit key** as in the Quick Extraction Guide (Android, step B).
6. **Iniciar** (start) and wait for the extraction-finished message. Do not close the tool midway: the acquisition would be left without a seal.

4. READ THE RESULT · Casos e Documentos > case > acquisition

Document / action	What it is
Relatório de Extração (Extraction Report)	The procedure: method, outcome of each module and hashes.
Relatório de Conteúdo (Content Report)	Messages, WhatsApp, contacts and photo locations, with the origin of each line (CSV + PDF).
Decifrar backup (Decrypt backup)	Handle WhatsApp with the 64-digit key you generated.
Conferir Integridade (Check Integrity)	Recomputes the hash of each file and compares it with the seal — practice this step too.

HOW TO READ EACH MODULE'S RESULT

Result	What it means
Concluído (Completed)	Obtained what it set out to.
Concluído com ressalva (Completed with caveat)	Obtained part and declared what it did not reach.
Não obteve dado (No data obtained)	Nothing was acquired. It does **not** mean the data does not exist on the device.

WHAT THE EMULATOR DOES NOT TEACH. Because it has root, the emulator shows extractions that **would not happen on a common phone without root**. Do not conclude that “it worked on the emulator” means “it works on device X”. The emulator trains the **operator** and the **workflow**; proof that a method works on a real model only comes from testing the real model.

Golden rule: what was not obtained is not what does not exist. The tool never asserts what it did not verify — and neither should you, while practicing.